

Element B: Documentation and Analysis of Prior Solution Attempts

Our school produces a lot of paper waste. A large majority of this waste comes from sheets of paper that have only been used or printed on one side. The other side is blank and completely usable, but the paper is getting thrown out anyway. This leads to more cost for the school since they have to pay more money for paper. It also led to more work for our housekeepers, who had to haul all of the extra paper waste out of the building and into dumpsters. The other paper waste comes from the composition notebooks that students use for classes. The cost of these notebooks comes directly out of each teacher's budget. However, in most cases, less than the first third of the notebook is used, and then it is not used again and just gets thrown out, even though the other two-thirds of the notebook has never been used.

One way to reuse paper waste involves applying nail polish remover (watered-down acetone) with a small brush or cotton swab to the ink on the paper and gently rubbing (- *Youtube*, n.d.). While this does remove the ink very effectively and quickly, it presents a risk of damaging the paper, and it can leave behind ink stains and warps in the paper.

Paper waste can be turned into new paper by mixing paper with water and blending the mix (Woodlark, 2022). After the mix dries, it will become usable paper. This presents a number of opportunities for aesthetic design with color, thickness, etc. However, the resulting paper tends to be flimsy, and it can have a rough surface. It was also relatively weak compared to normal paper.

Scrap paper can be stapled together and used as a notebook (Hesse, 2020). This provides a relatively easy, cheap solution, but came at the cost of time, and the paper was one-sided. It also does not yield a uniform result.

Our stakeholders' current solution is to throw away old paper, which is easy and cheap, but contributes to waste and forces our school to buy more new paper. Our stakeholders have made other attempts to solve this problem, one of which was having 8th-grade students collect paper waste to be recycled, but the 8th-grade students were too loud and interrupted classes. Since that attempt failed, our school has thrown everything out.



Nail Polish Remover Handmade Paper

Scrap Notebooks

References

- *YouTube*. (n.d.). Retrieved April 3, 2026, from

<https://www.youtube.com/watch?v=FgHPb0VsVp0&feature=youtu.be>

Woodlark. (2022, April 29). Diy handmade recycled paper—Woodlark blog.

Woodlark Blog - <https://woodlarkblog.com/diy-handmade-recycled-paper/>

Hesse, K. (2020, May 27). *DIY a scrap paper notepad—A journey to zero waste upcycling project • NourishNestBreathe*. NourishNestBreathe.

<https://nourishnestbreathe.com/scrap-paper-notepads/>